REVISION WORKSHEET

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وزارة التربية والتعليم

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Cycle 3 (10048) | Dibba Al Hisn | Sharjah Eastern

Practice Questions and Worked Answers

Grade 10 Advanced | Module 17 Lesson 1 | Interference

All 30 questions are followed by answers. Study the reasoning, formula, substitution and units. Equivalent scientifically correct wording is acceptable.

Useful equations and guidance: Double slit: $\Delta x = \lambda L/d$; $x_m = m\lambda L/d$. Constructive path difference = $m\lambda$; destructive = $(m + ½)\lambda$. Small-angle approximation applies.

Term 1 2026–2027 | Textbook alignment: uploaded Grade 10 book, printed pages 41–49.

P01. Coherent sources maintain:

A. Unrelated frequencies and phases B. A constant phase difference
C. Zero wavelength D. Zero intensity at all times

Answer

B. A constant phase difference.

P02. Constructive interference occurs when two equal-frequency waves arrive:

A. Always out of phase by half a cycle B. Without overlapping
C. In phase D. At different unrelated frequencies only

Answer

C. In phase.

P03. For equal-amplitude waves, destructive interference is strongest at phase difference:

A. 0° B. 360°
C. 720° D. 180°

Answer

D. 180°.

P04. In a double-slit experiment, increasing wavelength with other quantities fixed makes bright-fringe spacing:

A. Larger B. Smaller
C. Zero D. Unchanged

Answer

A. Larger.

P05. A central bright fringe in the ideal symmetric double-slit pattern corresponds to path difference:

A. One half wavelength B. Zero
C. An arbitrary nonzero distance D. One quarter wavelength always

Answer

B. Zero.

P06. In a double-slit experiment, λ = 500 nm, slit separation d = 0.250 mm and screen distance L = 2 m. Find adjacent bright-fringe spacing in millimetres.

A. 8 mm B. 2 mm
C. 4 mm D. -4 mm

Answer

C. 4 mm. $\Delta x = \lambda L/d = (500 \times 10^{-9})(2)/(0.250 \times 10^{-3}) = 0.004$ m = 4 mm.

P07. Double-slit fringe spacing is 4 mm at L = 2.4 m with d = 0.300 mm. Find the wavelength in nanometres.

A. 1000 nm B. 250 nm
C. -500 nm D. 500 nm

Answer

D. 500 nm. $\lambda = \Delta x\, d/L = (0.004)(0.300 \times 10^{-3})/2.4 = 5 \times 10^{-7}$ m = 500 nm.

P08. For λ = 600 nm, d = 0.300 mm and L = 1.00 m, find the distance of bright fringe order m = 2 from the central maximum, in millimetres.

A. 4 mm B. 8 mm
C. 2 mm D. -4 mm

Answer

A. 4 mm. $x_m = m\lambda L/d = 2(600 \times 10^{-9})(1.00)/(0.300 \times 10^{-3}) = 4$ mm.

P09. For wavelength 500 nm, calculate the path difference at the bright fringe of order m = 2.

A. 2000 nm B. 1000 nm
C. 500 nm D. -1000 nm

Answer

B. 1000 nm. Constructive interference requires $\Delta\text{path} = m\lambda = 2 \times 500 = 1000$ nm.

P10. For a film with one phase reversal, the minimum nonzero thickness for constructive reflection at normal incidence is t = λ/(4n). Find t for λ = 500 nm and n = 1.50.

A. 166.7 nm B. 41.67 nm
C. 83.33 nm D. -83.33 nm

Answer

C. 83.33 nm. $t = \lambda/(4n) = 500/(4 \times 1.50) = 83.33$ nm.

P11. Define interference.

Answer

The superposition of waves that produces an enhanced or reduced resultant disturbance depending on their relative phases.

P12. Define coherent light.

Answer

Light whose waves maintain a stable phase relationship, normally with the same frequency for a stable interference pattern.

P13. What is a bright interference fringe?

Answer

A region where waves combine constructively, giving relatively high intensity.

P14. Explain why ordinary independent lamps do not produce a stable visible fringe pattern easily.

Answer

Their relative phase changes rapidly, so interference averages out over typical observation times.

P15. Why can a soap film display colours in white light?

Answer

Reflections from its two surfaces interfere. Thickness, angle and wavelength determine which colours are enhanced or reduced.

P16. In a double-slit experiment, λ = 500 nm, slit separation d = 0.250 mm and screen distance L = 2.5 m. Find adjacent bright-fringe spacing in millimetres.

Answer

$\Delta x = \lambda L/d = (500 \times 10^{-9})(2.5)/(0.250 \times 10^{-3}) = 0.005$ m = 5 mm.

P17. In a double-slit experiment, λ = 500 nm, slit separation d = 0.250 mm and screen distance L = 3 m. Find adjacent bright-fringe spacing in millimetres.

Answer

$\Delta x = \lambda L/d = (500 \times 10^{-9})(3)/(0.250 \times 10^{-3}) = 0.006$ m = 6 mm.

P18. Double-slit fringe spacing is 6 mm at L = 3.6 m with d = 0.300 mm. Find the wavelength in nanometres.

Answer

$\lambda = \Delta x\, d/L = (0.006)(0.300 \times 10^{-3})/3.6 = 5 \times 10^{-7}$ m = 500 nm.

P19. Double-slit fringe spacing is 8 mm at L = 4.8 m with d = 0.300 mm. Find the wavelength in nanometres.

Answer

$\lambda = \Delta x\, d/L = (0.008)(0.300 \times 10^{-3})/4.8 = 5 \times 10^{-7}$ m = 500 nm.

P20. For λ = 600 nm, d = 0.300 mm and L = 1.00 m, find the distance of bright fringe order m = 3 from the central maximum, in millimetres.

Answer

$x_m = m\lambda L/d = 3(600 \times 10^{-9})(1.00)/(0.300 \times 10^{-3}) = 6$ mm.

P21. For λ = 600 nm, d = 0.300 mm and L = 1.00 m, find the distance of bright fringe order m = 4 from the central maximum, in millimetres.

Answer

$x_m = m\lambda L/d = 4(600 \times 10^{-9})(1.00)/(0.300 \times 10^{-3}) = 8$ mm.

P22. For wavelength 500 nm, calculate the path difference at the bright fringe of order m = 3.

Answer

Constructive interference requires $\Delta\text{path} = m\lambda = 3 \times 500 = 1500$ nm.

P23. For wavelength 500 nm, calculate the path difference at the bright fringe of order m = 4.

Answer

Constructive interference requires $\Delta\text{path} = m\lambda = 4 \times 500 = 2000$ nm.

P24. For a film with one phase reversal, the minimum nonzero thickness for constructive reflection at normal incidence is $t = \lambda/(4n)$. Find t for $\lambda = 550$ nm and $n = 1.50$.

Answer

$t = \lambda/(4n) = 550/(4 \times 1.50) = 91.67$ nm.

P25. For a film with one phase reversal, the minimum nonzero thickness for constructive reflection at normal incidence is $t = \lambda/(4n)$. Find t for $\lambda = 600$ nm and $n = 1.50$.

Answer

$t = \lambda/(4n) = 600/(4 \times 1.50) = 100$ nm.

P26. In a double-slit experiment, predict the fringe-spacing change if screen distance doubles while all else stays fixed.

Answer

From $\Delta x = \lambda L/d$, fringe spacing doubles. The small-angle approximation must remain appropriate.

P27. If slit separation doubles with the same wavelength and screen distance, what happens to fringe spacing?

Answer

Spacing halves because Δx is inversely proportional to d.

P28. Why does a white-light double-slit pattern have a white central fringe and coloured neighbouring fringes?

Answer

At the centre all wavelengths have zero path difference and reinforce. Away from the centre their maxima occur at different positions, producing coloured fringes.

P29. A student applies the one-phase-reversal thin-film formula without checking surrounding media. Evaluate the procedure.

Answer

It is incomplete. Whether reflection reverses phase depends on the relative refractive indices at each boundary. The correct constructive/destructive condition must reflect those phase reversals.

P30. Design a method to determine wavelength with a double slit.

Answer

Measure d and L, measure a span covering several fringes to estimate mean spacing Δx, and use $\lambda = \Delta x\, d/L$. Repeat readings and use consistent units.